Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

MLOps Experimental Learning Assignment:

End-to-End ML Model Development, CI/CD, and Production Deployment Experimental Learning.

Objective:

Design, develop, and deploy a scalable, reproducible machine learning solution using modern MLOps best practices. The assignment emphasizes practical automation, experiment tracking, CI/CD pipelines, containerization, cloud deployment, and monitoring, mirroring real-world production scenarios.

Dataset: Title: Heart Disease UCI Dataset

- Source: UCI Machine Learning Repository
  - https://archive.ics.uci.edu/ml/datasets/heart+Disease
- CSV containing 14+ features (age, sex, blood pressure, cholesterol, etc.) and a binary target (presence/absence of heart disease).

Problem Statement:

Build a machine learning classifier to predict the risk of heart disease based on patient health data, and deploy the solution as a cloud-ready, monitored API.

Assignment Tasks:

1. Data Acquisition & Exploratory Data Analysis (EDA) [5 marks]

- Obtain the dataset (provide download script or instructions). Clean and preprocess the data (handle missing values, encode features). Perform EDA with professional visualizations (histograms, correlation heatmaps, class balance).

2. Feature Engineering & Model Development [8 marks]

- Prepare final ML features (scaling, encoding). Build and train at least two classification models (e.g., Logistic Regression and Random Forest). Document model selection and tuning process. Evaluate using cross validation and relevant metrics (accuracy, precision, recall, ROC-AUC).

3. Experiment Tracking [5 marks]

- Integrate MLflow (or similar) for experiment tracking. Log parameters, metrics, artifacts, and plots for all runs.

4. Model Packaging & Reproducibility [7 marks]

- Save the final model in a reusable format (e.g., MLflow, pickle, ONNX). Write a clean requirements.txt (or Conda env file). Provide preprocessing pipeline/transformers to ensure full reproducibility.

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

5. CI/CD Pipeline & Automated Testing [8 marks]

- Write unit tests for data processing and model code (Pytest or unit test). Create a GitHub Actions (or Jenkins) pipeline: Linting, unit test, and model training steps. Artifacts/logging for each workflow run.

6. Model Containerization [5 marks]

- Build a Docker container for the model serving API (Flask or FastAPI recommended). Expose /predict endpoint, accept JSON input, return prediction and confidence. Container must build and run locally with sample input.

7. Production Deployment [7 marks]

- Deploy the Dockerized API to a public cloud or local Kubernetes (GKE, EKS, AKS, or Minikube/Docker Desktop). Use a deployment manifest or Helm chart. Expose via Load Balancer or Ingress. Verify endpoints and provide deployment screenshots.

8. Monitoring & Logging [3 marks]

- Integrate logging of API requests. Demonstrate simple monitoring (Prometheus + Grafana or API metrics/logs dashboard).

9. Documentation & Reporting [2 marks]

- Submit a professional Markdown or PDF report including:
- Setup/install instructions.
- EDA and modelling choices.
- Experiment tracking summary.
- Architecture diagram.
- CI/CD and deployment workflow screenshots.
- Link to code repository.

Assignment Implementation:

Assignment Objective

This assignment requires the development of a complete MLOps workflow for a machine learning problem. The objective is not only to build a predictive model, but also to show how the solution can be prepared for repeatable training, automated testing, containerized deployment, monitoring, and documentation. The repository delivered here covers all of those aspects for a heart disease prediction system using the Cleveland heart disease dataset.

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Requirement 1: Data acquisition and exploratory data analysis

For this requirement focuses on obtaining the dataset, understanding its structure, and performing exploratory analysis before modelling.

Following things has been done,

- The raw dataset is stored in **data/raw/heart_cleveland.csv** and is loaded through the reusable preprocessing module in **src/data/preprocessing.py**

- The workflow includes a data quality report with missing value analysis, duplicate checks, feature statistics, and target distribution review.

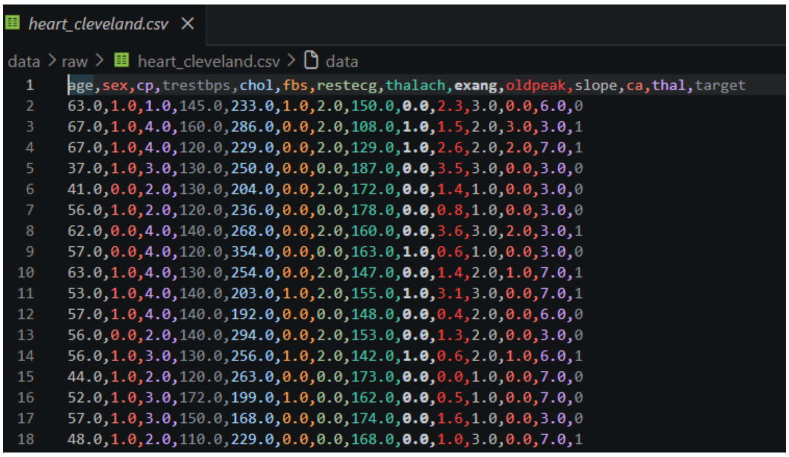
- Missing values are handled systematically using median imputation for numeric features and mode imputation for categorical features.

- The repository also includes notebook-based exploratory work in notebooks/01_data_acquisition_eda.ipynb

- The solution shows how raw data is loaded, checked, cleaned, and inspected before any model is trained.

- The process is documented in code and supported by visual evidence.

Visual evidence:

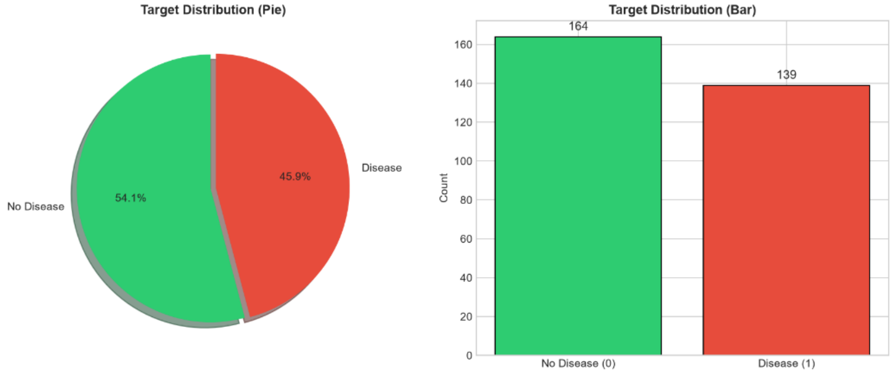


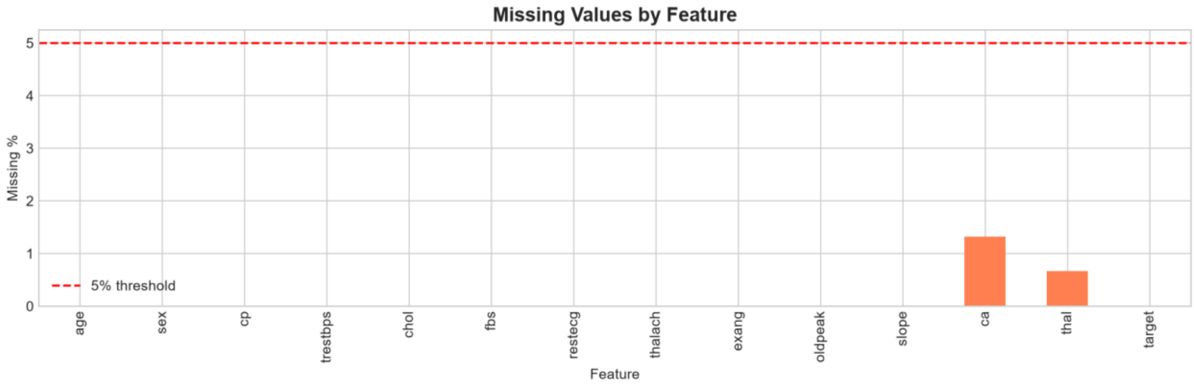
```
heart_cleveland.csv
data > raw > heart_cleveland.csv > data
age,sex,cp,trestbps,chol,fbs,restecg,thalach,exang,oldpeak,slope,ca,thal,target
63.0,1.0,1.0,145.0,233.0,1.0,2.0,150.0,0.0,2.3,3.0,0.0,6.0,0
67.0,1.0,4.0,160.0,286.0,0.0,2.0,108.0,1.0,1.5,2.0,3.0,3.0,1
67.0,1.0,4.0,120.0,229.0,0.0,2.0,129.0,1.0,2.6,2.0,2.0,7.0,1
37.0,1.0,3.0,130.0,250.0,0.0,0.0,187.0,0.0,3.5,3.0,0.0,3.0,0
41.0,0.0,2.0,130.0,204.0,0.0,2.0,172.0,0.0,1.4,1.0,0.0,3.0,0
56.0,1.0,2.0,120.0,236.0,0.0,0.0,178.0,0.0,0.8,1.0,0.0,3.0,0
62.0,0.0,4.0,140.0,268.0,0.0,2.0,160.0,0.0,3.6,3.0,2.0,3.0,1
57.0,0.0,4.0,120.0,354.0,0.0,0.0,163.0,1.0,0.6,1.0,0.0,3.0,0
63.0,1.0,4.0,130.0,254.0,0.0,2.0,147.0,0.0,1.4,2.0,1.0,7.0,1
53.0,1.0,4.0,140.0,203.0,1.0,2.0,155.0,1.0,3.1,3.0,0.0,7.0,1
57.0,1.0,4.0,140.0,192.0,0.0,0.0,148.0,0.0,0.4,2.0,0.0,6.0,0
56.0,0.0,2.0,140.0,294.0,0.0,2.0,153.0,0.0,1.3,2.0,0.0,3.0,0
56.0,1.0,3.0,130.0,256.0,1.0,2.0,142.0,1.0,0.6,2.0,1.0,6.0,1
44.0,1.0,2.0,120.0,263.0,0.0,0.0,173.0,0.0,0.0,1.0,0.0,7.0,0
52.0,1.0,3.0,172.0,199.0,1.0,0.0,162.0,0.0,0.5,1.0,0.0,7.0,0
57.0,1.0,3.0,150.0,168.0,0.0,0.0,174.0,0.0,1.6,1.0,0.0,3.0,0
48.0,1.0,2.0,110.0,229.0,0.0,0.0,168.0,0.0,1.0,3.0,0.0,7.0,1
```

RAW DATA

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

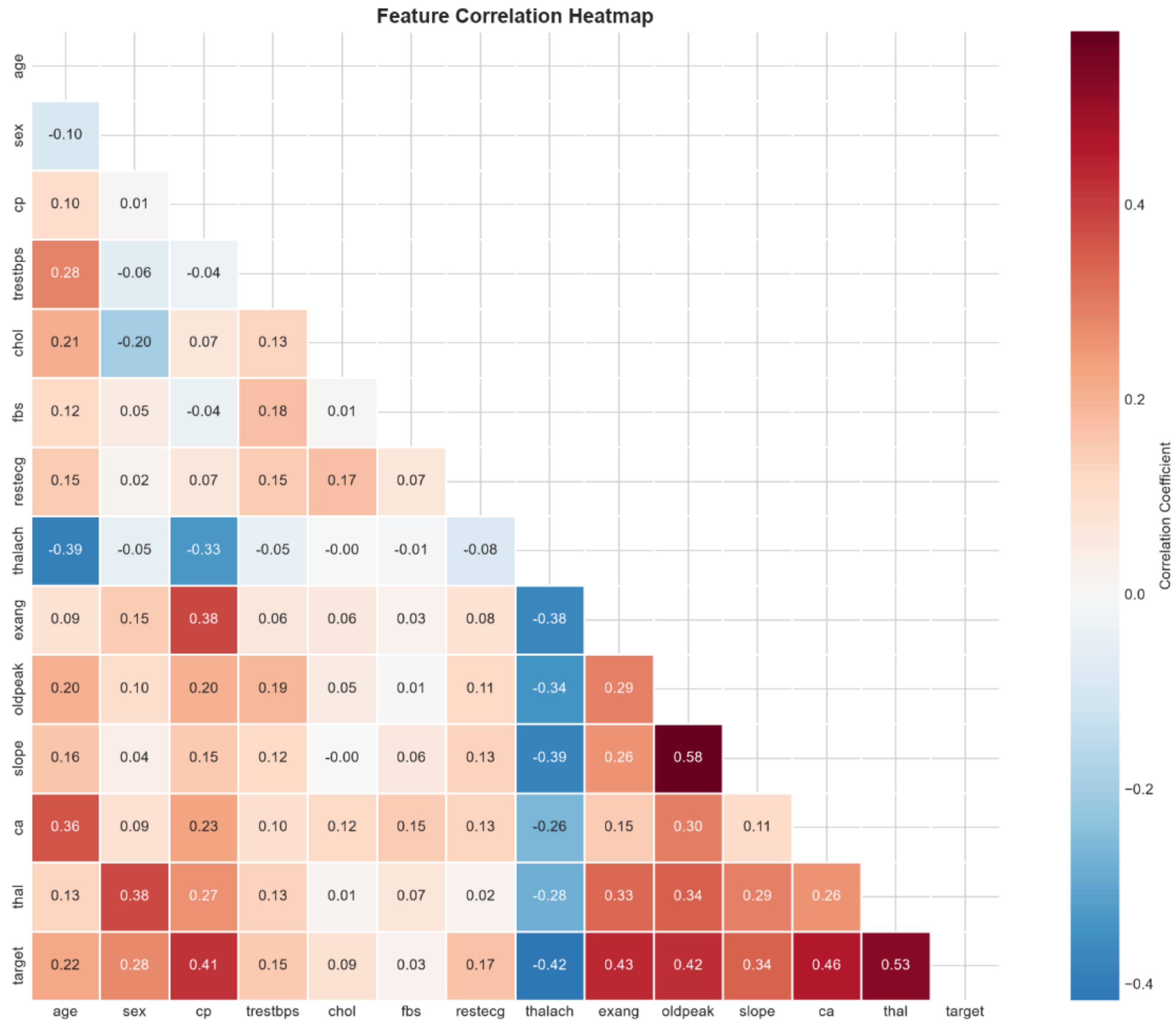
Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN





Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN



Relevant files in Repository:

- src/data/preprocessing.py
- data/raw/heart_cleveland.csv
- notebooks/01_data_acquisition_eda.ipynb
- screenshots/target_distribution.png
- screenshots/missing_values.png
- screenshots/correlation_heatmap.png

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Requirement 2: Feature engineering and model development

- The preprocessing pipeline defines numeric and categorical features and applies scaling and imputation in **src/data/preprocessing.py**

- The training workflow in **src/models/train.py** trains multiple classifiers, including logistic regression, random forest, gradient boosting, and support vector machine.

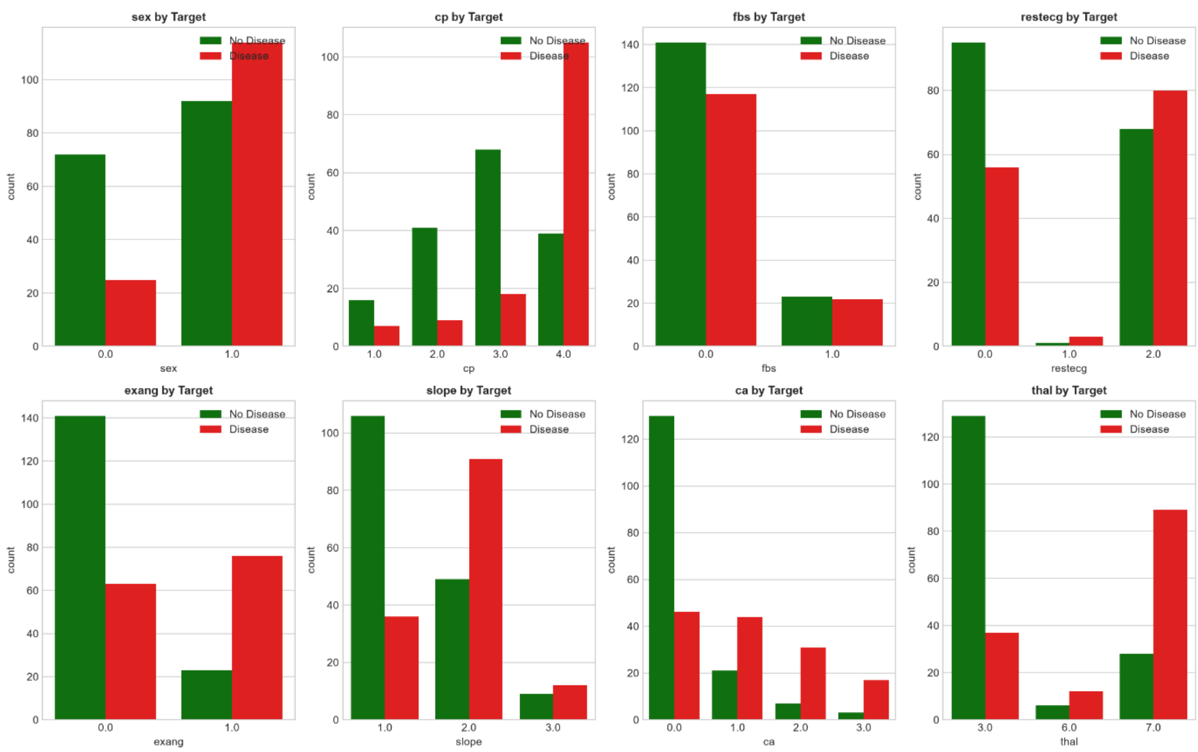
- Each model is evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

- The best-performing model is saved for later inference.

- The project demonstrates model experimentation rather than a single fixed classifier.

- The model development process is reusable and reproducible.

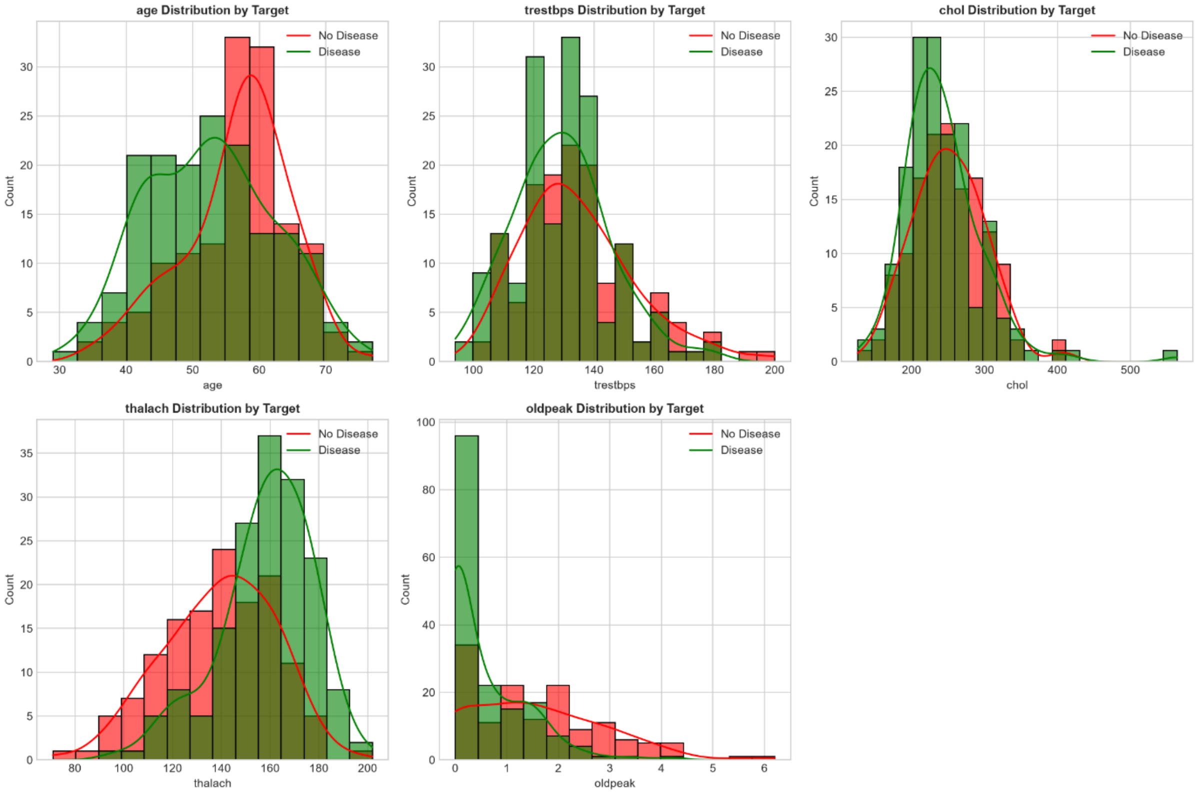
Visual evidence:



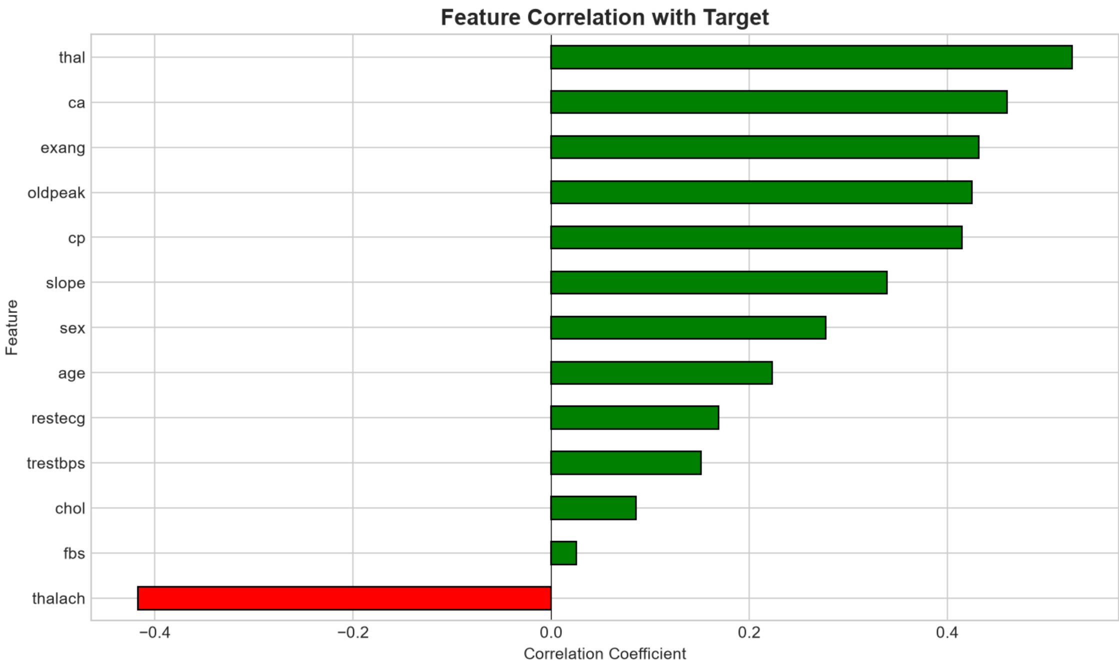
categorical_distributions

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

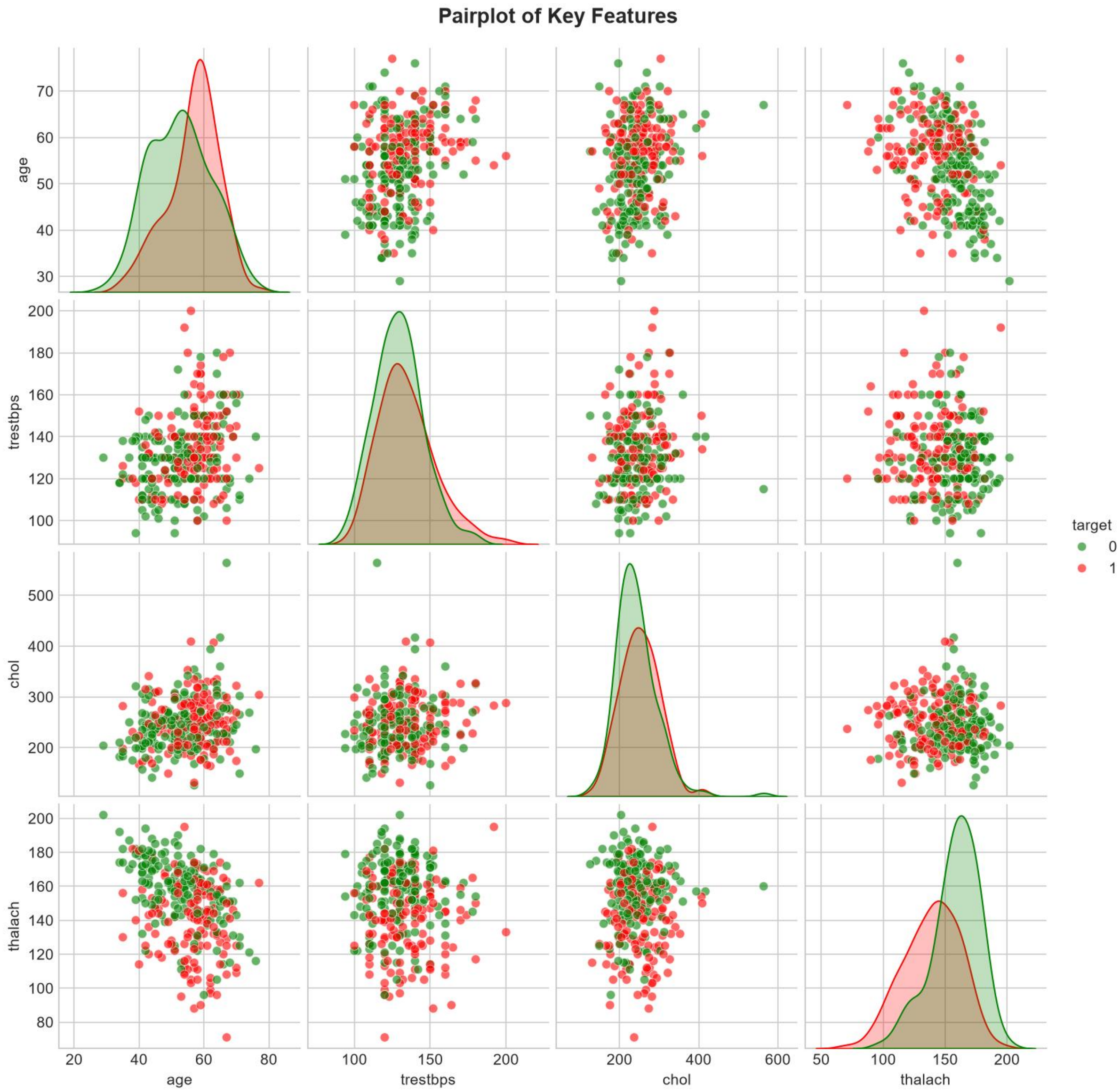


numeric_distributions



Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

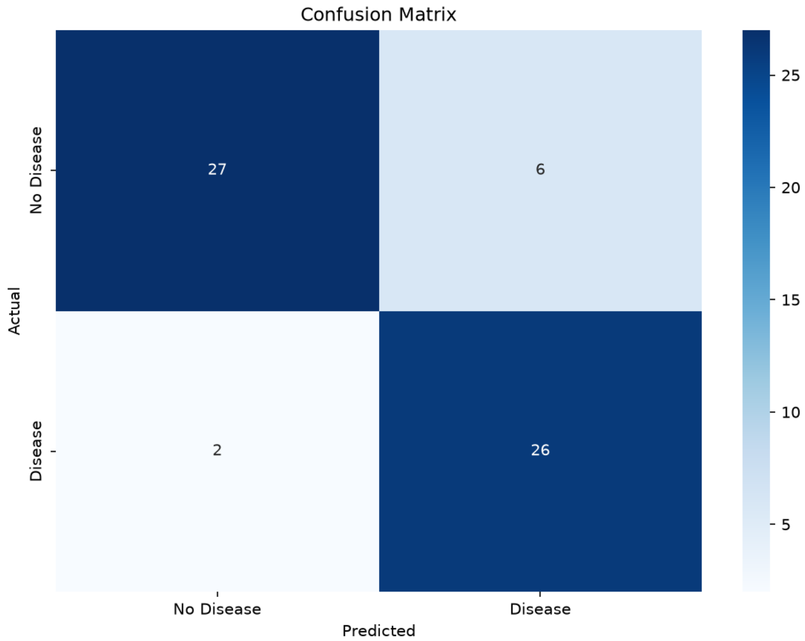
Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN



Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

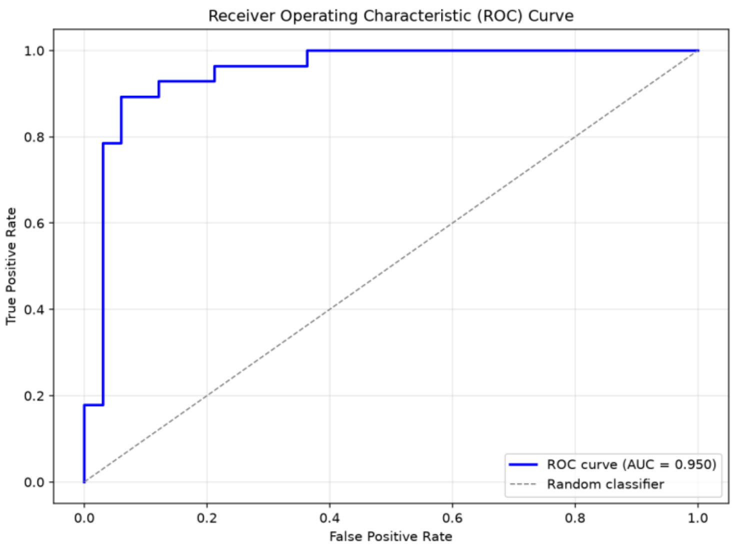
Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Logistic regression



screenshots > logistic_regression > classification_report.txt

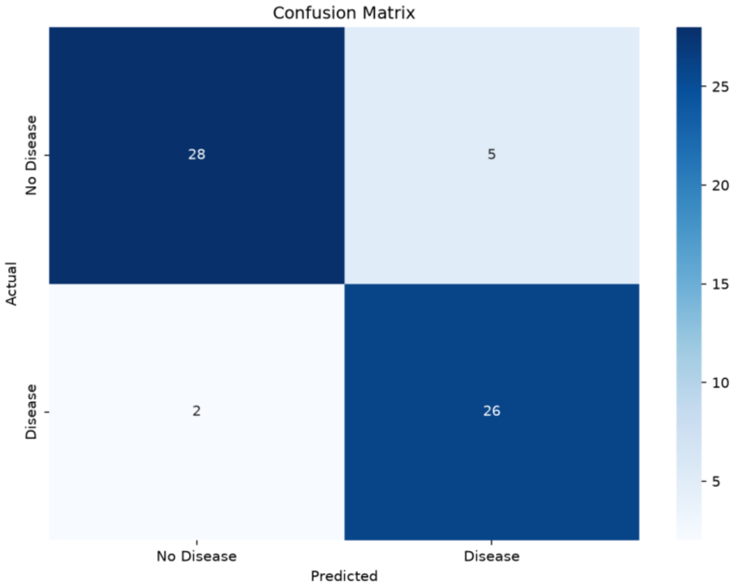
	precision	recall	f1-score	support
0	0.93	0.82	0.87	33
1	0.81	0.93	0.87	28
accuracy			0.87	61
macro avg	0.87	0.87	0.87	61
weighted avg	0.88	0.87	0.87	61



Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

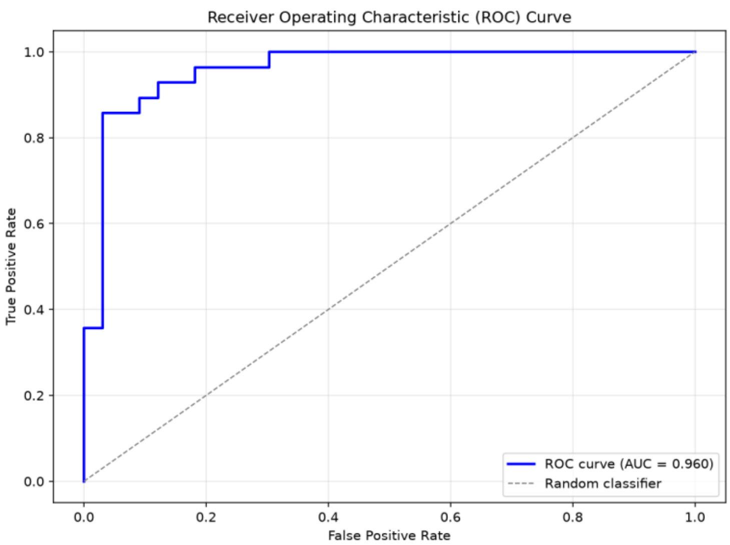
Random forest



classification_report.txt

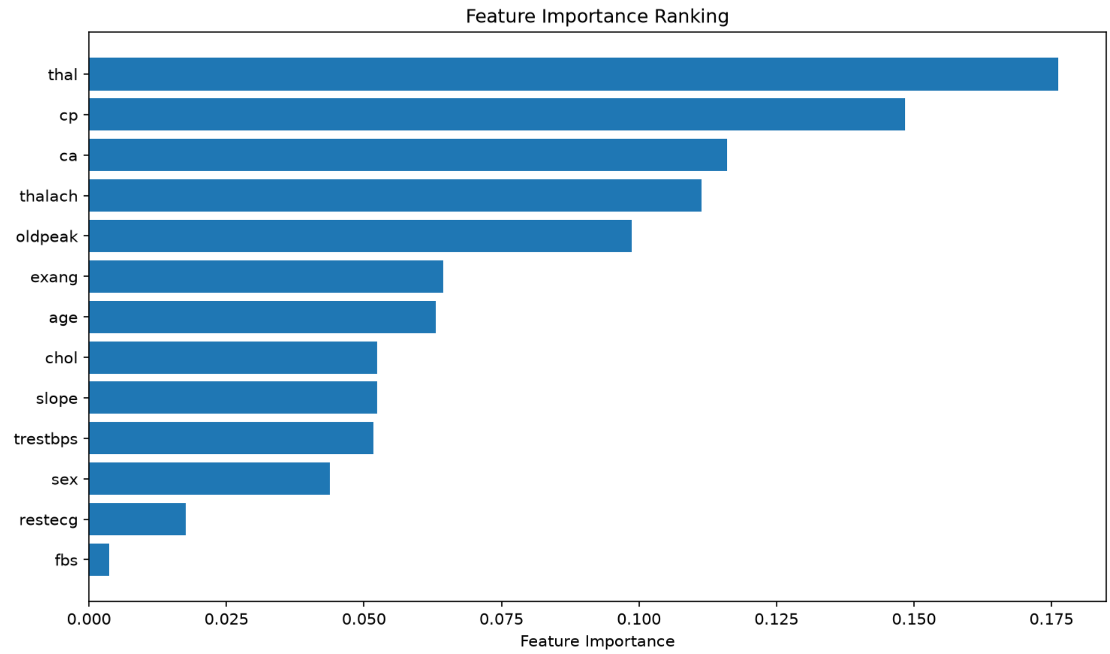
screenshots > random_forest > classification_report.txt

	precision	recall	f1-score	support
0	0.93	0.85	0.89	33
1	0.84	0.93	0.88	28
accuracy			0.89	61
macro avg	0.89	0.89	0.89	61
weighted avg	0.89	0.89	0.89	61

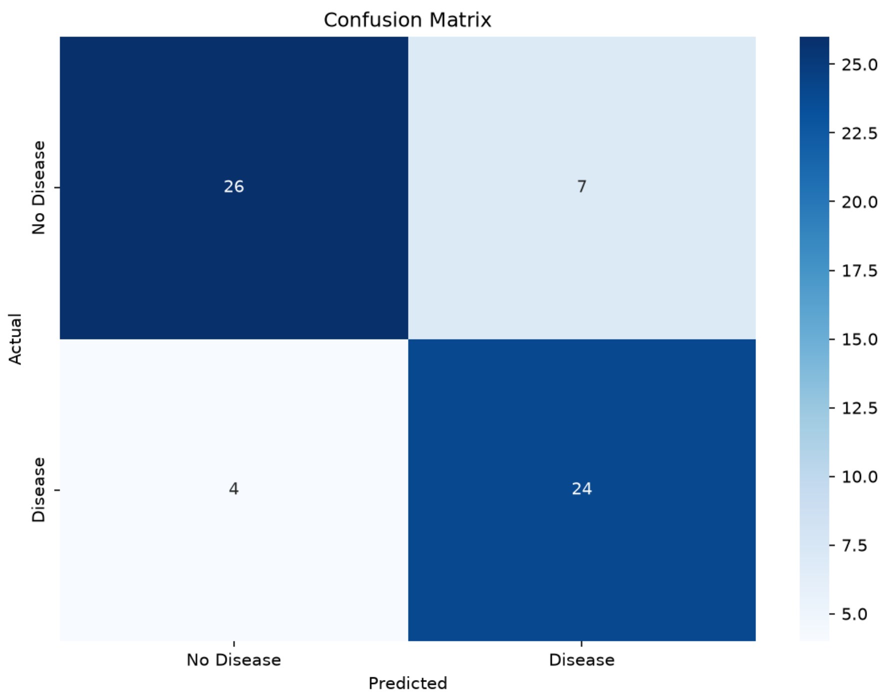


Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN



Support Vector Machines (SVM)



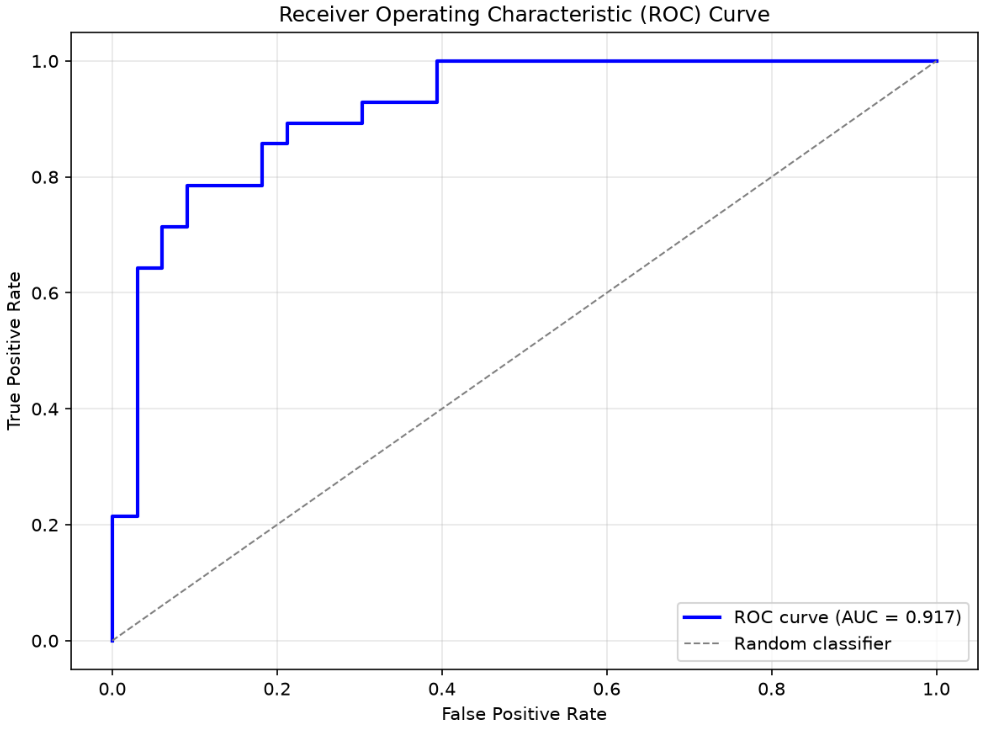
Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

classification_report.txt

screenshots > svm > classification_report.txt

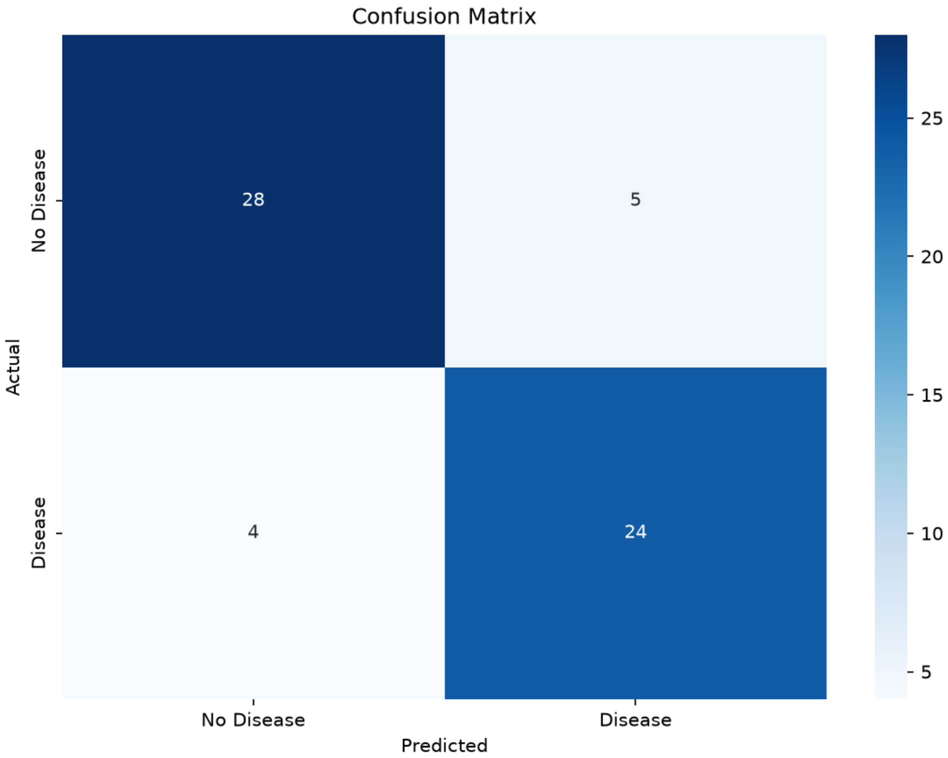
	precision	recall	f1-score	support
0	0.87	0.79	0.83	33
1	0.77	0.86	0.81	28
accuracy			0.82	61
macro avg	0.82	0.82	0.82	61
weighted avg	0.82	0.82	0.82	61



Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Gradient boosting:



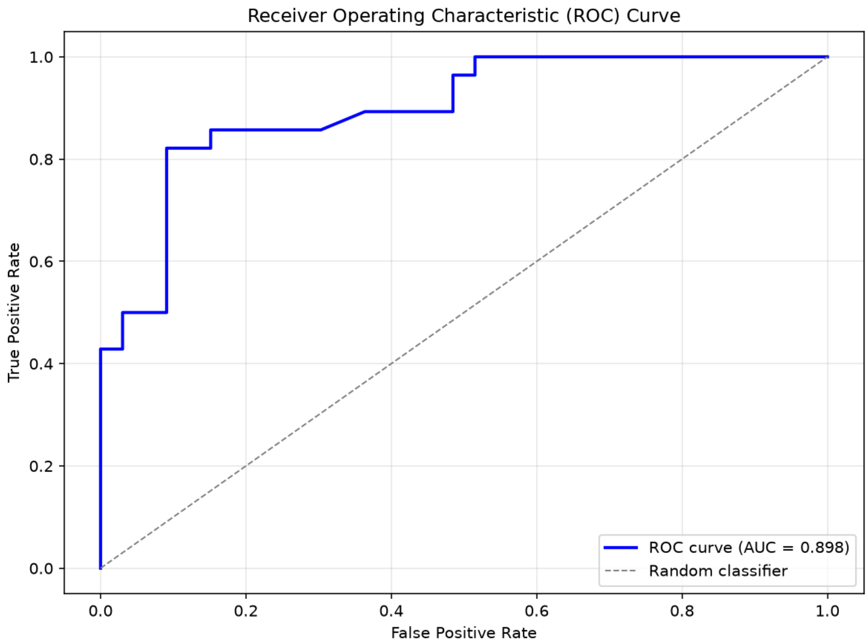
classification_report.txt

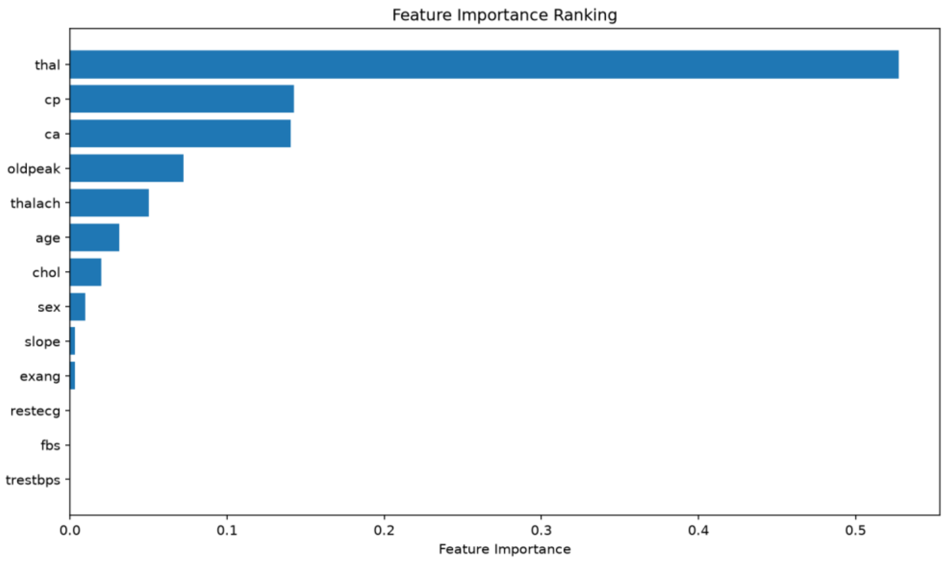
screenshots > gradient_boosting > classification_report.txt

	precision	recall	f1-score	support
0	0.88	0.85	0.86	33
1	0.83	0.86	0.84	28
accuracy			0.85	61
macro avg	0.85	0.85	0.85	61
weighted avg	0.85	0.85	0.85	61

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN





Relevant files in Repository:

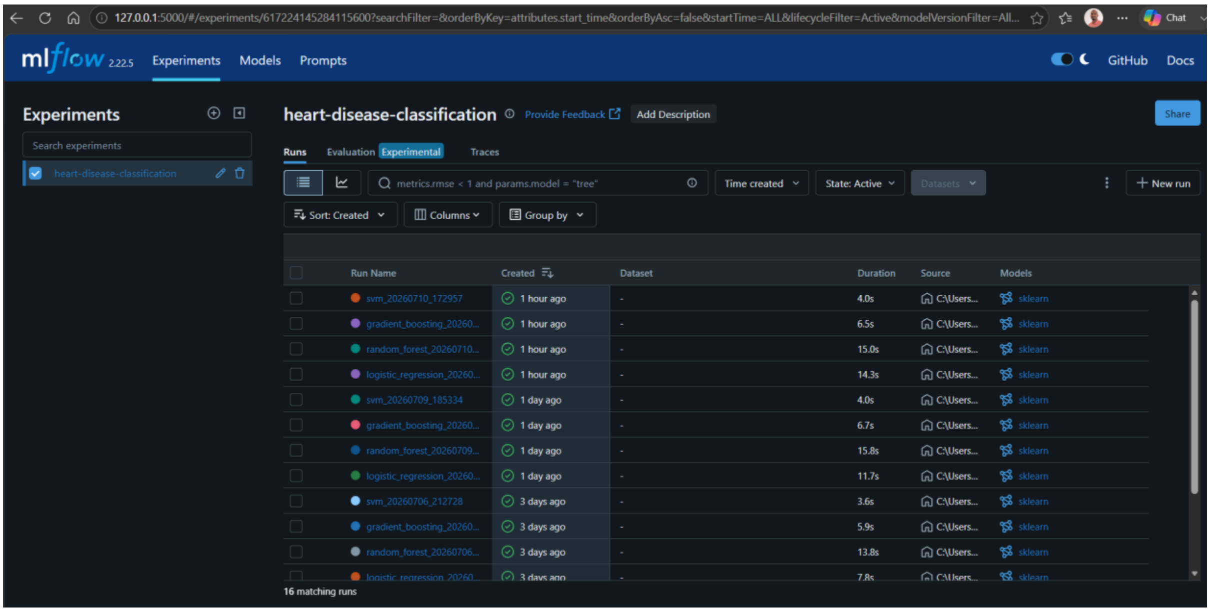
- src/data/preprocessing.py
- src/models/train.py
- models/final_model.joblib
- screenshots/categorical_distributions.png
- screenshots/numeric_distributions.png
- screenshots/target_correlation.png

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Requirement 3: Experiment tracking with MLflow

- The training script uses MLflow to log experiment metadata, model parameters, evaluation metrics, and generated plots.
- The local MLflow tracking server can be used to compare runs and inspect artifacts.
- Experiment artifacts are stored under mlruns
- Teams can reproduce and compare model runs over time.
- The reporting and performance comparison become traceable and auditable.



Relevant files in Repository:

- src/models/train.py
- mlruns
- notebooks/02_model_training.ipynb

Requirement 4: Model packaging and reproducibility

- The trained model and fitted preprocessor are stored as joblib artifacts in **models/final_model.joblib** and **models/preprocessor.joblib**
- The code is organized so that data processing, model training, and prediction serving can be run independently.
- Dependency management is handled through **requirements.txt** and **setup.py**
- The trained assets can be reused for prediction without retraining.
- The environment is documented and can be recreated on another machine.

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Relevant files in Repository:

- models/final_model.joblib
- models/preprocessor.joblib
- requirements.txt
- setup.py

Requirement 5: CI/CD and automated testing

- Automated unit tests are included in **tests/test_api.py**, **tests/test_data_processing.py**, and **tests/test_model.py**.

- A GitHub Actions workflow is defined in **.github/workflows/ci-cd.yml** to run linting, tests, model training, and Docker build steps.

- The workflow is structured to provide reproducible validation each time the code changes.

- The repository includes automated checks that help catch regressions early.

- The workflow reflects a realistic CI/CD approach expected in MLOps assignments.

Verification evidence:

- If we ran the following command in the workspace:

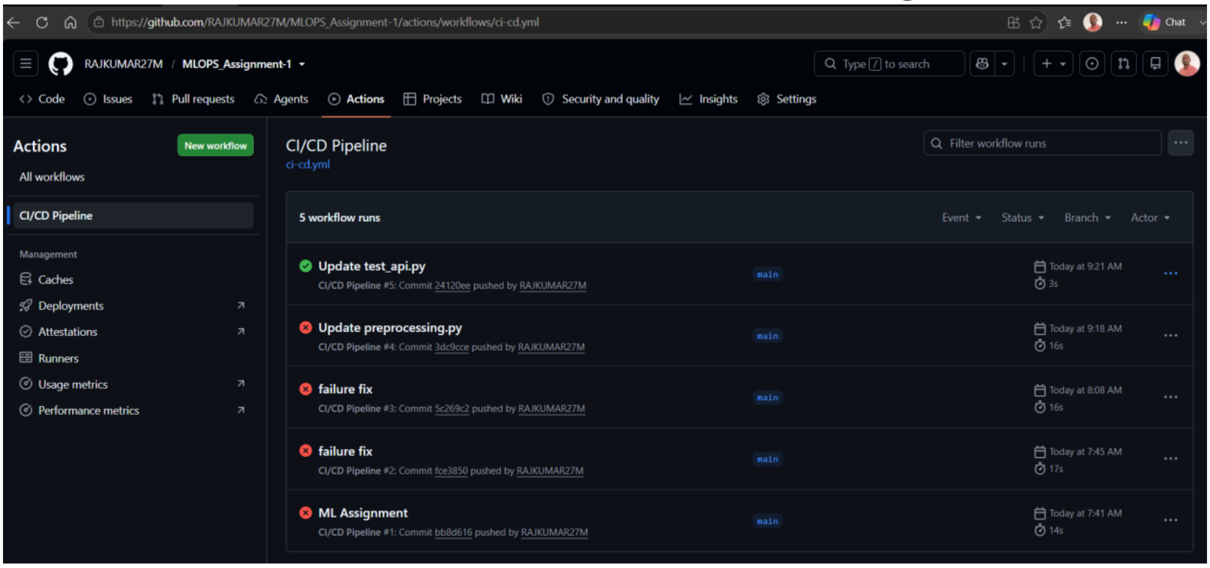
pytest -q

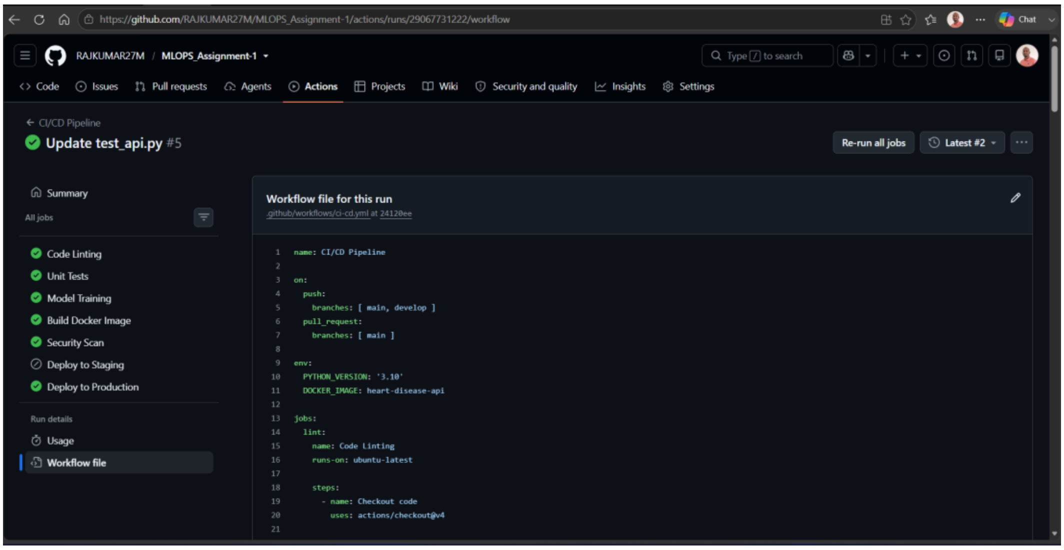
Result:

```
PS C:\Users\Rajkumar Madhavan\OneDrive\Documents\GitHub\MLOPS_Assignment-1> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& "c:\Users\R
ajkumar Madhavan\OneDrive\Documents\GitHub\MLOPS_Assignment-1\.venv\Scripts\Activate.ps1")
(.venv) PS C:\Users\Rajkumar Madhavan\OneDrive\Documents\GitHub\MLOPS_Assignment-1> pytest -q
==================================================================== test session starts ====================================================================
platform win32 -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\Rajkumar Madhavan\OneDrive\Documents\GitHub\MLOPS_Assignment-1
configfile: pytest.ini
testpaths: tests
plugins: anyio-4.14.1, cov-7.1.0
collected 60 items

tests\test_api.py ........................                                                                                                             [ 40%]
tests\test_data_processing.py ...................                                                                                                      [ 71%]
tests\test_model.py .................                                                                                                                  [100%]

===================================================================== warnings summary =====================================================================
..\..\..\Desktop\MLOPS\MLOPS-Assignment\.venv\Lib\site-packages\fastapi\testclient.py:1
  C:\Users\Rajkumar Madhavan\OneDrive\Desktop\MLOPS\MLOPS-Assignment\.venv\Lib\site-packages\fastapi\testclient.py:1: StarletteDeprecationWarning: Using `httpx
` with `starlette.testclient` is deprecated; install `httpx2` instead.
    from starlette.testclient import TestClient as TestClient  # noqa

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
============================================================== 60 passed, 1 warning in 17.41s ==============================================================
```

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN





Relevant files in Repository:

- .github/workflows/ci-cd.yml
- tests/test_api.py
- tests/test_data_processing.py
- tests/test_model.py

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Requirement 6: Model containerization

This requirement expects the application to be packaged in a portable container image.

- The container definition is provided in **Dockerfile.**

- The Docker image installs dependencies, copies the application code, and starts the API service.

- Local orchestration and service composition are supported through **docker-compose.yml**

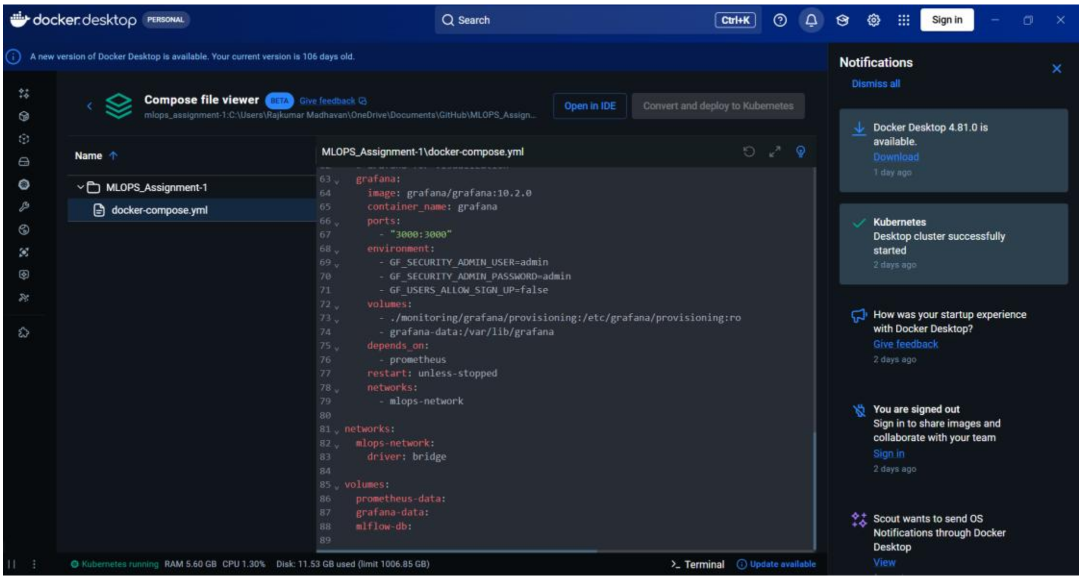
- The model API can be built and run as an isolated environment, which is essential for deployment consistency.

Relevant files in Repository:

- Dockerfile

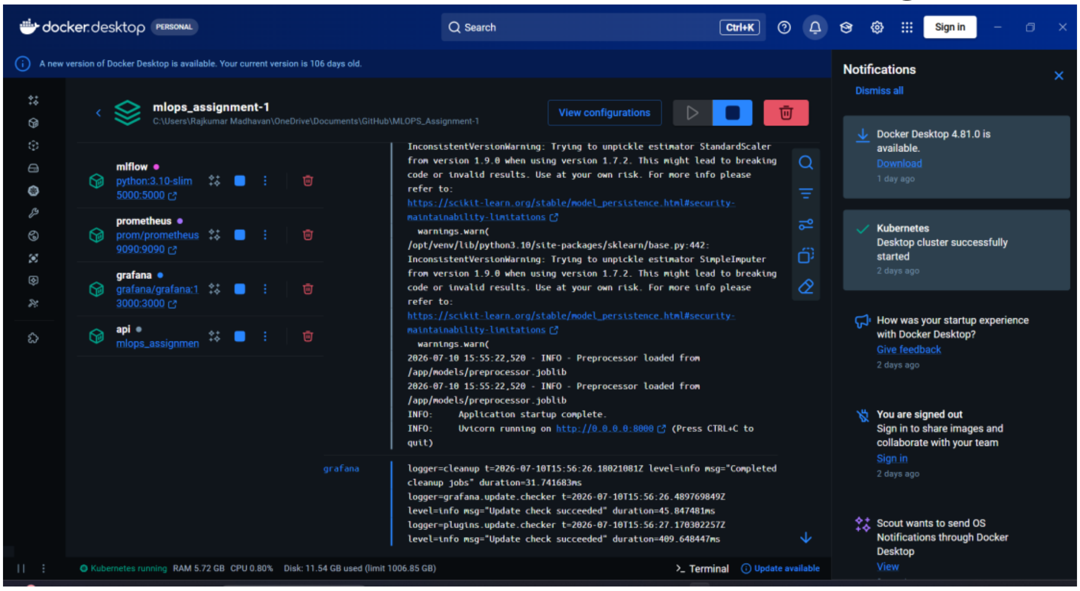
- docker-co

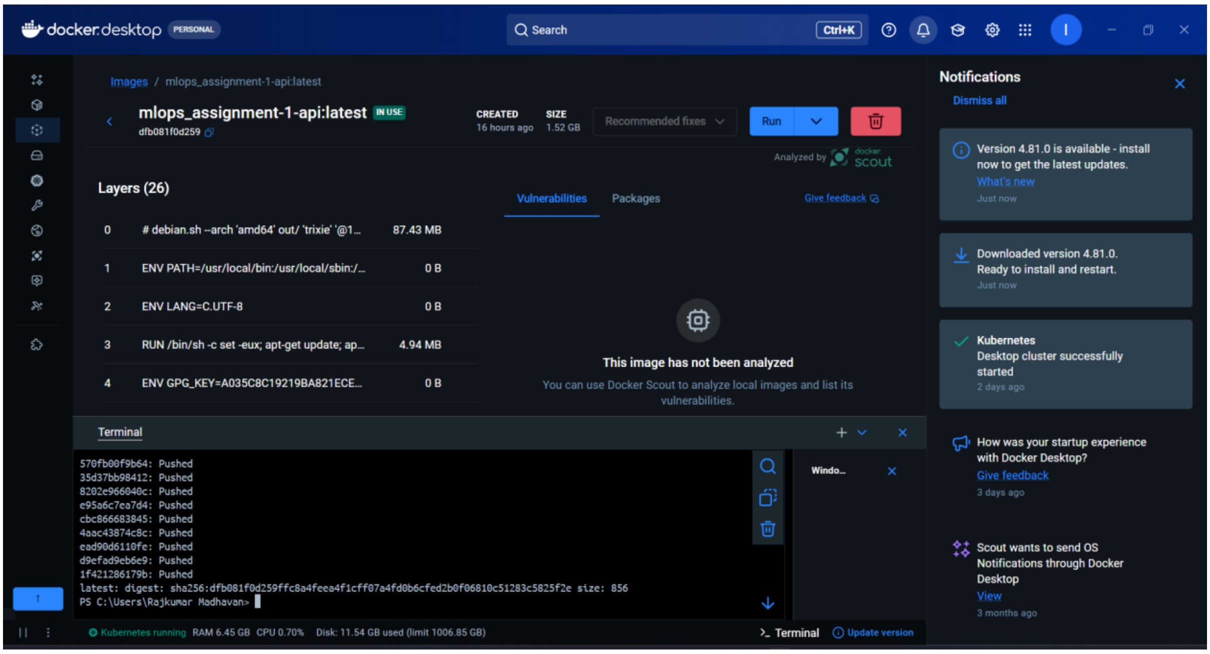
mpose.yml



Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN





Requirement 7: Production-style deployment

- Kubernetes deployment manifests are included in **deployment/kubernetes/deployment.yaml**.

- A Helm chart is also provided in **deployment/helm/heart-disease-api.**

- The deployment uses a rolling update strategy, readiness and liveness probes, and resource requests and limits.

- The project demonstrates how the API could be served in a container orchestration environment, not only as a local script.

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Relevant files in Repository:

- deployment/kubernetes/deployment.yaml

- deployment/kubernetes/service.yaml

- deployment/kubernetes/ingress.yaml

- deployment/helm/heart-disease-api

Requirement 8: Monitoring and logging

- The API in **src/api/main.py** includes request logging and prediction endpoints.

- Prometheus metrics are exposed through the application and configured in **monitoring/prometheus/prometheus.yml.**

- Grafana provisioning and dashboard configuration are available in **monitoring/grafana**.

- The application is prepared for production observability and can be monitored for usage and performance.

Relevant files in Repository:

- src/api/main.py

- monitoring/prometheus/prometheus.yml

- monitoring/grafana

Requirement 9: Documentation and reporting

- The main project documentation is available in **README.md**

- This detailed report is provided in **FINAL_REPORT.md**

- A shorter submission summary is available in **GITHUB_SUBMISSION_SUMMARY.md**

- The repository also preserves screenshots and artifacts in **screenshots**

Code Repository Link

GitHub repository: https://github.com/RAJKUMAR27M/MLOPS_Assignment-1

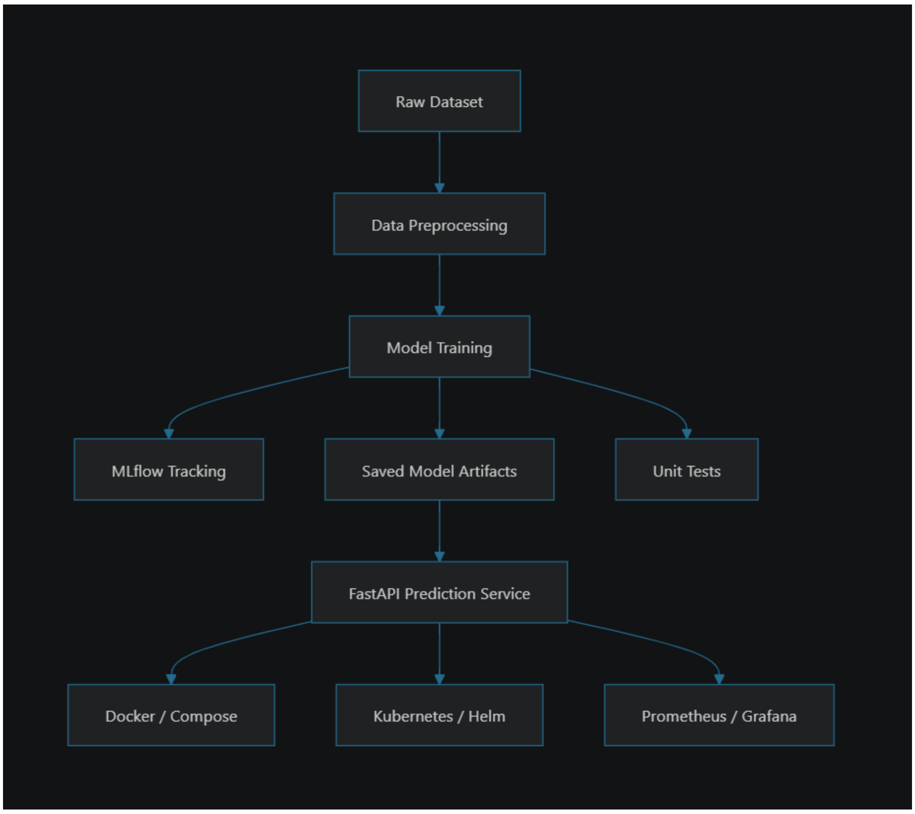
Link to Video: https://drive.google.com/file/d/1IjVZdUONCeYiAnmgSpUvEuxINhC50K-R/view?usp=sharing

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Project Architecture Summary

The repository is organized in a way that reflects a realistic MLOps pipeline:



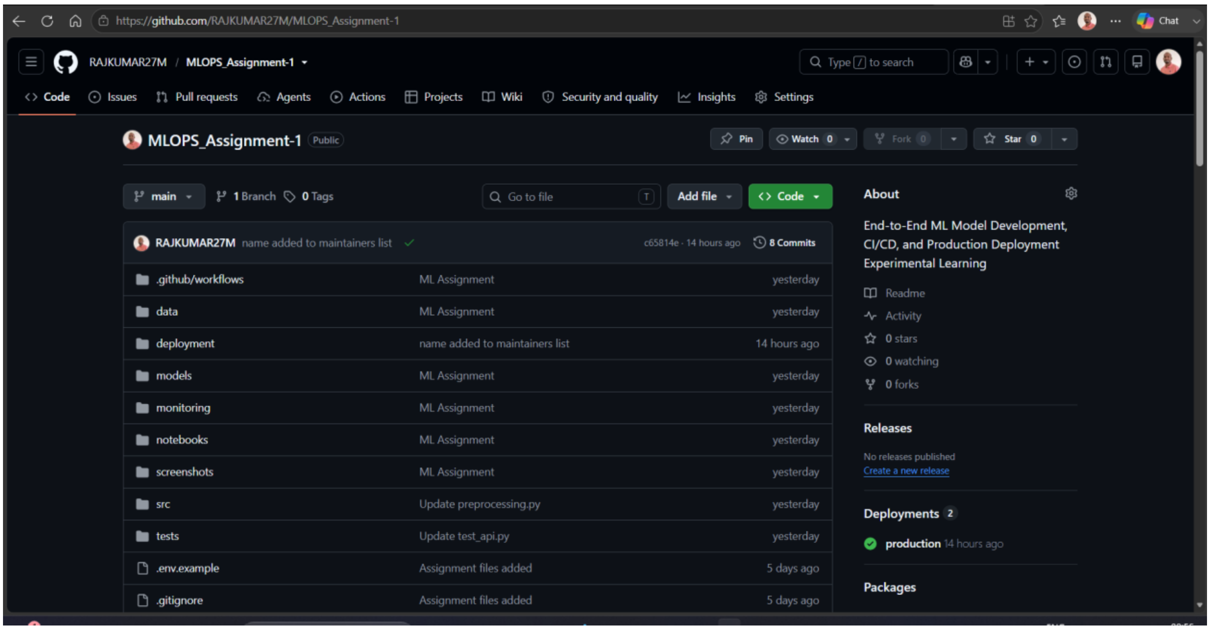
- Data layer: **src/data**
- Model training layer: **src/models**
- Serving layer: **src/api**
- Testing layer: **tests**
- Deployment and orchestration: **deployment**
- Monitoring: **monitoring**

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Code Repository Link

GitHub repository: https://github.com/RAJKUMAR27M/MLOPS_Assignment-1



How to Run the Project

- **Prerequisites**

  - Install Python 3.11+

  - Install Git and VS Code

  - Ensure We are in the project root folder

- **Create and activate the environment**

  Open PowerShell in the project folder and run below commands one by one:

  - python -m venv .venv
  - .\.venv\Scripts\Activate.ps1
  - python -m pip install --upgrade pip
  - pip install -r requirements.txt

- **Explore the dataset and understand the problem**

  Review the raw data file at **data/raw/heart_cleveland.csv**

  Open the notebooks in **notebooks** for EDA and preprocessing understanding

  Inspect the preprocessing logic in **src/data/preprocessing.py**

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

- **Run preprocessing and prepare the data**

  Run below command:

  ```
  python -m src.data.preprocessing
  ```

This step loads the data, handles missing values, encodes features, and prepares training data.

- **Train the machine learning models**

  Run below commands:

  ```
  python -m src.models.train
  ```

This trains multiple classifiers, evaluates them, and saves the best model plus preprocessor to **models.**

- **Track experiments with MLflow**

  Start the MLflow UI:

  ```
  python -m mlflow ui --host 127.0.0.1 --port 5000
  ```

  Then open:

  http://127.0.0.1:5000

We can compare training runs and inspect metrics/artifacts stored in **mlruns.**

- **Run the automated tests**

  Run below command:

  ```
  pytest -q
  ```

This verifies preprocessing, training, and API behaviour.

- **Start the prediction API locally**

  Run below command:

  ```
  python -m uvicorn src.api.main:app --host 127.0.0.1 --port 8000
  ```

  Then open:

  http://127.0.0.1:8000/docs

We can test the prediction endpoint from the Swagger UI.

- **Containerize the application**

  Build the Docker image:

  Run below command in powershell but make sure docker running

  ```
  docker build -t heart-disease-api .
  ```

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

- **Run the container:**

  docker run -p 8000:8000 heart-disease-api

- **Run with Docker Compose**

  Use:

  docker-compose up

This starts the services defined in **docker-compose.yml**

- **Deploy with Kubernetes or Helm**

Deployment assets are available in **deployment/kubernetes** and **deployment/helm/heart-disease-api**

Example:

In powershell

kubectl apply -f deployment/kubernetes/

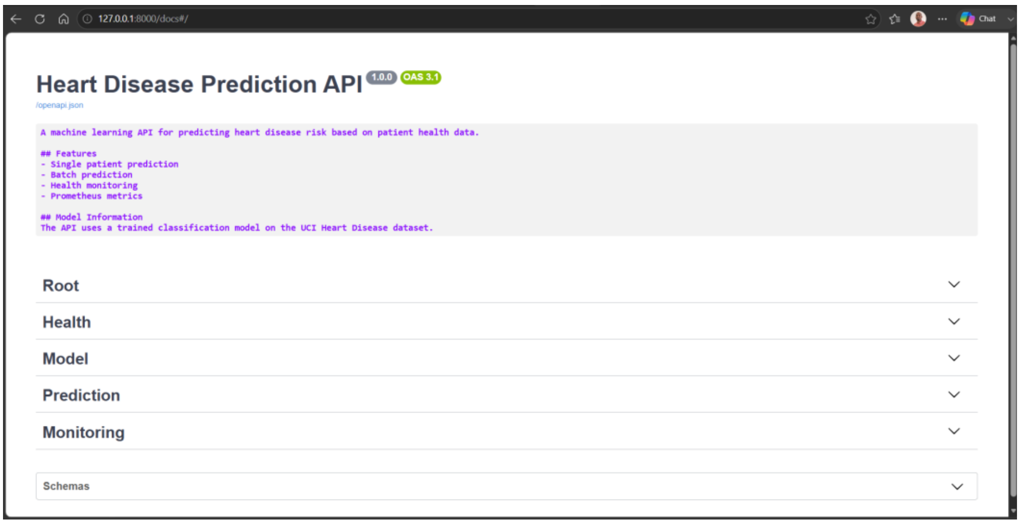
- **Monitor the application**

Monitoring configuration is available in **monitoring/prometheus** and **monitoring/grafana**

- Prometheus scrapes the metrics endpoint

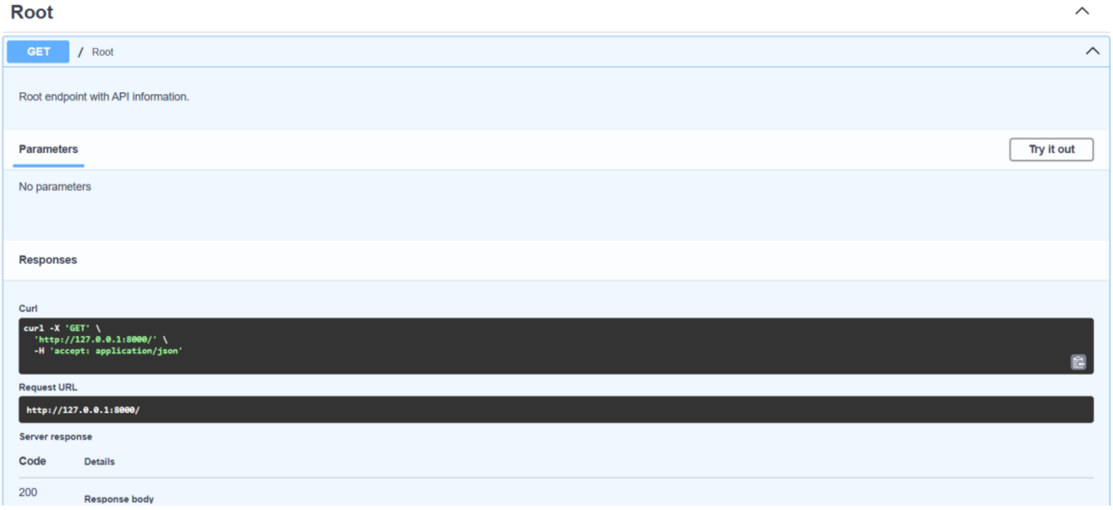
- Grafana visualizes the service metrics

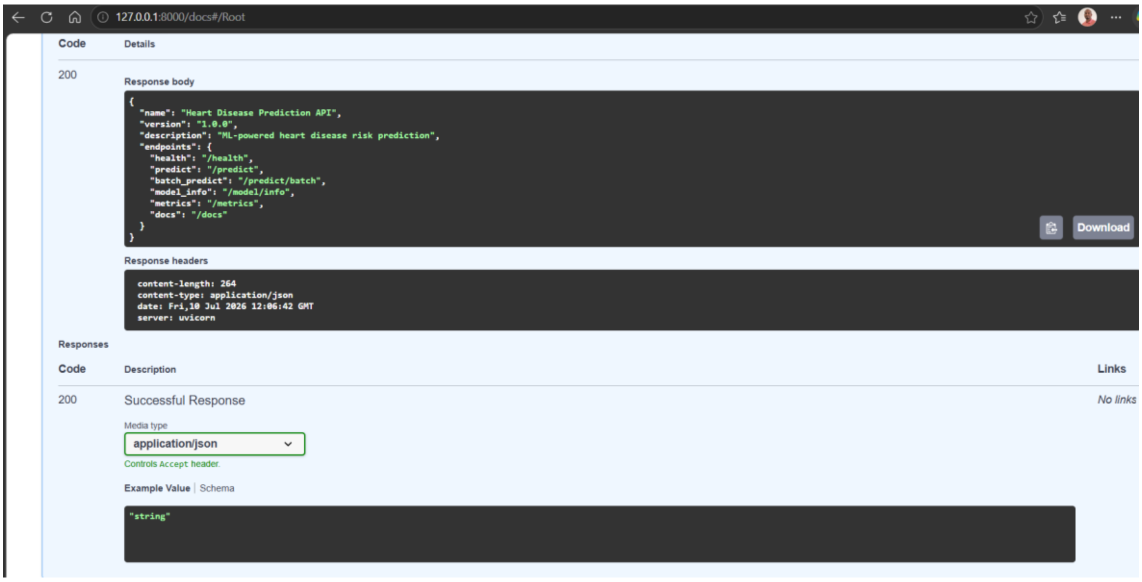
Swagger UI Screenshots:

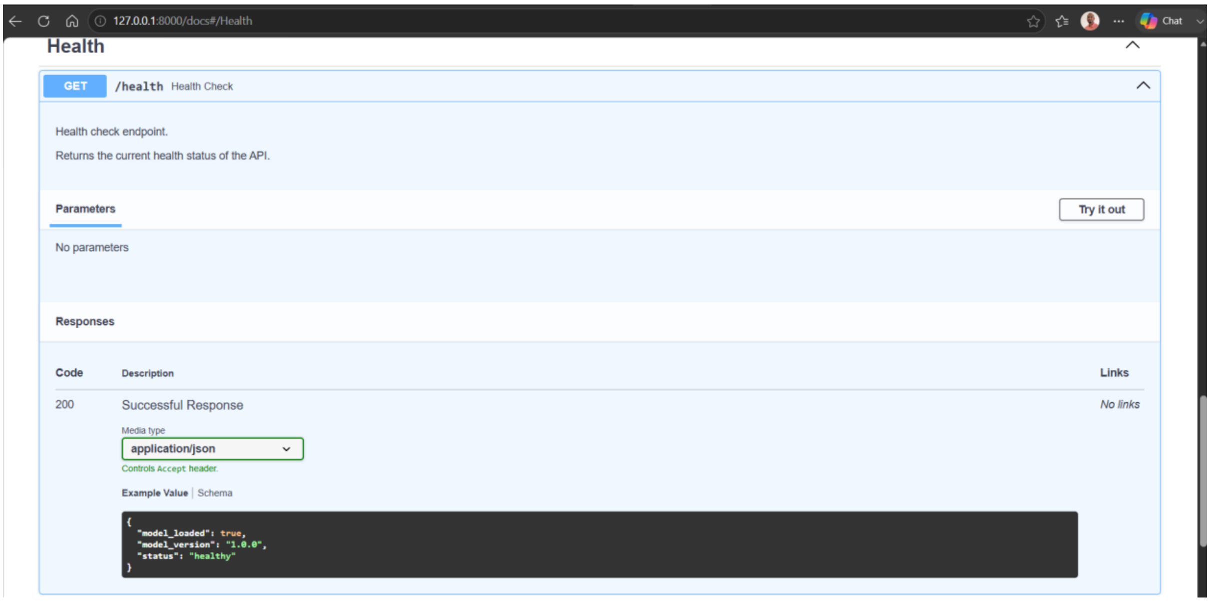


Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN







Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Model

GET /model/info Get Model Info

Get model information.

Returns details about the loaded model.

Parameters Try it out

No parameters

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/model/info' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/model/info
```

Request URL

```
http://127.0.0.1:8000/model/info
```

Server response

Code	Details
200	Response body

```
{
  "model_name": "random_forest",
  "model_version": "1.0.0",
  "features": [
    "age",
    "sex",
    "cp",
    "trestbps",
    "chol",
    "fbs",
    "restecg",
    "thalach",
    "exang",
    "oldpeak",
    "slope",
    "ca",
    "thal"
  ],
  "created_at": "2026-07-10T17:30:01.177718"
}
```

Download

Response headers

```
content-length: 208
content-type: application/json
date: Fri,10 Jul 2026 12:07:15 GMT
server: uvicorn
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "created_at": "2024-01-15T10:30:00",
  "features": [
    "age",
    "sex",
    "cp",
    "trestbps",
    "chol",
    "fbs",
    "restecg",
    "thalach",
    "exang",
    "oldpeak",
    "slope",
    "ca",
    "thal"
  ],
  "model_name": "random_forest",
  "model_version": "1.0.0"
}
```

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Prediction

POST `/predict` Predict

Predict heart disease risk for a single patient.

- **age**: Age in years (20-100)
- **sex**: Sex (1 = male, 0 = female)
- **cp**: Chest pain type (0-3)
- **trestbps**: Resting blood pressure (mm Hg)
- **chol**: Serum cholesterol (mg/dl)
- **fbs**: Fasting blood sugar > 120 mg/dl (1 = true, 0 = false)
- **restecg**: Resting ECG results (0-2)
- **thalach**: Maximum heart rate achieved
- **exang**: Exercise induced angina (1 = yes, 0 = no)
- **oldpeak**: ST depression induced by exercise
- **slope**: Slope of peak exercise ST segment (0-2)
- **ca**: Number of major vessels (0-4)
- **thal**: Thalassemia (0-3)

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "age": 63,
  "ca": 0,
  "chol": 233,
  "cp": 3,
  "exang": 0,
  "fbs": 1,
  "oldpeak": 2.3,
  "restecg": 0,
  "sex": 1,
  "slope": 0,
  "thal": 1,
  "thalach": 150,
  "trestbps": 145
}
```

Responses

Code	Description	Links
200	Successful Response	*No links*

Media type

application/json

Controls `Accept` header.

Example Value | Schema

```
{
  "confidence": 0.85,
  "model_version": "1.0.0",
  "prediction": 1,
  "probability": 0.85,
  "risk_level": "High"
}
```

Code	Description	Links
422	Validation Error	*No links*

Media type

application/json

Example Value | Schema

```
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN

Responses

Code	Description	Links
200	Successful Response	*No links*
422	Validation Error	*No links*

200 Successful Response

Media type: application/json

Controls `Accept` header.

Example Value | Schema


```json
{
  "predictions": [
    {
      "confidence": 0.85,
      "model_version": "1.0.0",
      "prediction": 1,
      "probability": 0.85,
      "risk_level": "High"
    }
  ],
  "count": 0
}
```


422 Validation Error

Media type: application/json

Example Value | Schema


```json
{
  "detail": [
    {
      "loc": [
        "string",
        0
      ],
      "msg": "string",
      "type": "string",
      "input": "string",
      "ctx": {}
    }
  ]
}
```


Monitoring

GET `/metrics` Metrics

Prometheus metrics endpoint.

Returns metrics in Prometheus format for scraping.

Parameters — Try it out

No parameters

Responses

Curl

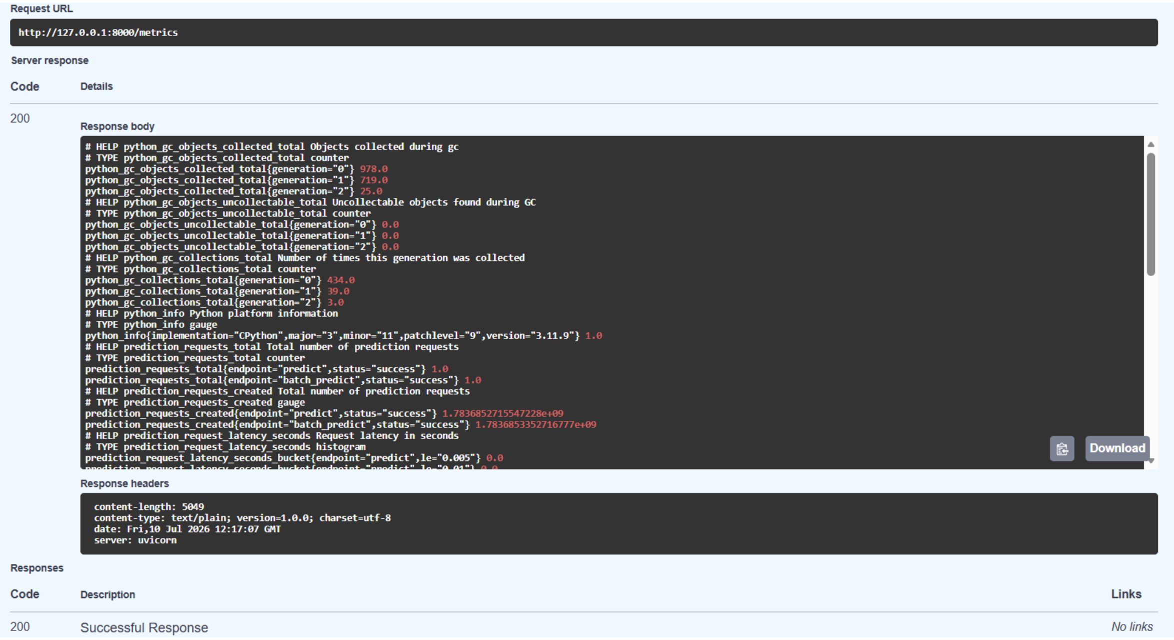
```
curl -X 'GET' \
  'http://127.0.0.1:8000/metrics' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/metrics
```

Machine Learning Operations (MLOps) AIMLCZG523 Assignment 01

Name: RAJ KUMAR M
BITS ID: 2024AD05110
Course: MTECH AI & ML
Email: 2024AD05110@WILP.BITS-PILANI.AC.IN



API deployed Publicly:

We can access it through public API URL https://mlops-assignment-1-api-latest.onrender.com/

Interactive API Docs: https://mlops-assignment-1-api-latest.onrender.com/docs

Conclusion

This project successfully demonstrates an end-to-end MLOps solution for heart disease prediction. It includes data preparation, model training, experiment tracking, testing, packaging, containerization, deployment assets, monitoring, and documentation. The repository is structured in a way that is suitable for both academic submission and practical industrial-style deployment demonstration.

Code Repository Link

GitHub repository: **https://github.com/RAJKUMAR27M/MLOPS_Assignment-1**

public API URL https://mlops-assignment-1-api-latest.onrender.com/

Interactive API Docs: https://mlops-assignment-1-api-latest.onrender.com/docs

Link to Video: **https://drive.google.com/file/d/1IjVZdUONCeYiAnmgSpUvEuxINhC50K-R/view?usp=sharing**